

Nucleon scattering analysis with a lane-consistent dispersive optical potential for Hf, W and Ta isotopes

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Abstract

Nucleon scattering on tungsten and neighboring nuclei are analyzed with a coupled-channels method based on a soft-rotator structure model. The multiple band couplings and nuclear stretching factors are built using nuclear wave functions of the soft-rotator model with the Hamiltonian parameters adjusted to reproduce the energy of low-lying collective levels of corresponding nuclei. A regional lane-consistent dispersive coupled-channels optical-model potential is derived to reproduce the neutron total cross sections, nucleon elastic and inelastic scattering angular distributions, and analyzing powers. Excellent agreement with measurements is obtained for neutron total cross sections of $^{182,183,184,186}\text{W}$, ^{178}Hf , and ^{181}Ta targets in the whole energy range from 100 keV to 200 MeV; other scattering data are found to be in good agreement with measurements. A deformation-dependent nuclear radius correction arising from the volume conservation condition is introduced and allowed to remove phenomenological radii dependencies on nuclear mass from the potential geometry.

Keywords: nucleon scattering, soft-rotator model, dispersive optical-model potential, coupled channels

(Some figures may appear in colour only in the online journal)

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1. Introduction

Tungsten isotopes are structural nuclei that are statically well deformed and are thought to be structurally similar to deformed heavy fission-product and actinide nuclei. Nuclear data of tungsten play an important role in nuclear engineering applications as well as in geophysics and astrophysics. Tungsten is an important structural material in fusion devices, being a strong candidate as the plasma-facing component material in magnetic confinement fusion reactors [1, 2] due to a high melting point and high sputtering resistance to energetic particles. In the Chinese Initiative Accelerator Driven Systems project, tungsten is also considered to be a candidate for spallation targets and other structural materials [3]. Tungsten is also used in high-temperature reactors [4]. In addition, nuclear data of tungsten isotopes are related to determination of the age and origin of the Earth, Moon and chondrules [5, 6]. However, there are still significant differences among existing nuclear data evaluations for tungsten isotopes in different evaluated nuclear data libraries. A reduction of their uncertainties for elastic and inelastic scattering data on tungsten isotopes has initiated additional work on the improvement of nuclear data modeling and evaluation.

The tungsten isotopes are at the rotational edge of the $A = 190$ – 200 transitional region, where even and odd nuclei change shapes (from prolate at the low- A end to oblate at the high- A end) [7]. They offer an stringent testbed for investigations of nuclear structure and the effects of deformations. However, up to now, many nucleon–nucleus scattering analyses on tungsten isotopes have been based on the spherical optical model. Quite a few analyses based on the coupled-channel approach can be found [8–10], in which the rigid rotor model is usually used by considering the coupling of only a few levels (usually three to five) of the axially symmetric ground-state (GS) rotational band. It was pointed out that optical model calculations with a number of coupled levels lower than seven on even–even targets are inadequate to give proper compound cross-section values [11].

Recently, some authors [12] have developed a model that extended Tamura’s [13] coupling formalism to consider low-lying collective bands of a vibrational nature observed in deformed nuclei. This model introduces ‘effective’ deformations—matrix elements of nuclear deformation operators—that define coupling arising from changes to a nucleus shape. Both even–even and odd- A actinides were analyzed in reference [12], where the ‘effective’ deformations were treated as fitting parameters. However, for odd- A nuclei, only excited bands based on single-particle excitations were taken into account.

Later, a soft-rotator nuclear structure model was used [14, 15] to calculate the ‘effective’ deformations for even–even actinides. It proved the importance of multi-band coupling, nucleus stretching and dynamic volume conservation effects for the calculation of optical-model cross sections in deformed nuclides. Finally, this approach was extended in reference [16] to describe nucleon scattering on odd- A nuclides, by calculating the odd- A nucleus ‘effective’ deformations from the corresponding even–even core. As a result, vibrational and GS bands built on the same single-particle state can be coupled for odd- A targets.

This work is devoted to a new analysis of nucleon scattering on tungsten isotopes and neighboring nuclei by the coupled-channels method, in which the nuclear structure effects are properly considered, i.e. more low-lying collective levels of different bands will be included into consideration. Then, we derive a regional lane-consistent dispersive coupled-channels optical-model potential (OMP) to consistently describe nucleon scattering both on even–even and odd- A targets. A joint analysis is carried out using almost all types of experimental nucleon–nucleus scattering data, including cross sections, elastic and inelastic angular distributions, and analyzing powers for both even–even and odd- A targets. It is expected that such an analysis can

Table 1. SRM Hamiltonian parameters and static deformation parameters used to reproduce the experimental level schemes of $^{182,184,186}\text{W}$, ^{178}Hf , and ^{180}Hf . A detailed explanation of SRM Hamiltonian parameters can be found in references [14, 15]. All Hamiltonian parameters are dimensionless (except $\hbar\omega_0$ which is given in MeV).

Target	$\hbar\omega_0$ (MeV)	$\mu_{\beta_{20}}$	μ_{γ_0}	γ_0	a_{32}	a_{42}	$\mu_{\epsilon}/\beta_{30}$	δ_n
^{182}W	1.0368	0.27876	0.27997	0.24768	0.002	0.026	2.5	15.495
^{184}W	0.99150	0.27597	0.28356	0.27094	0.002	0.026	2.5	14.495
^{186}W	0.94456	0.28077	0.32009	0.2972	0.002	0.026	2.5	12.493
^{178}Hf	1.0432	0.22125	0.33827	0.2585	0.256	0.026	0.47	15.211
^{180}Hf	1.0769	0.24768	0.27118	0.2352	0.002	0.0	2.5	12.493

Table 2. Energies of collective excited levels ordered by rotational bands for selected isotopes, used for SRM parameter determination and in coupling schemes. Level energies are available in ENSDF evaluated nuclear structure data [19–23] (unit is MeV).

I^π	^{182}W	^{184}W	^{186}W	^{178}Hf	^{180}Hf
GS rotational band with positive parity					
2^+	0.1001	0.1112	0.1225	0.0932	0.0933
4^+	0.3294	0.3640	0.3965	0.3066	0.3086
6^+	0.6805	0.7483	0.8093	0.6322	0.6408
8^+	0.1144	1.2522	1.3492	1.0585	1.0840
β -rotational band with positive parity					
0_1^+	1.1358	1.0025	0.8836	1.1994	1.1019
2_1^+	1.2574	1.1214	1.0302	1.2767	1.1834
4_1^+	1.5102	1.3604	1.2989	1.4504	1.3696
Nonaxial rotational band with positive parity					
2_2^+	1.2214	0.9033	0.7378	1.1746	1.1997
3_2^+	1.3311	1.0059	0.8623	1.2685	1.2911
4_2^+	1.4428	1.1338	1.0067	1.3845	1.4093
5_2^+	1.6235	1.2949	1.1973	1.5331	1.5573
γ -rotational band with positive parity					
0_3^+		1.3222	1.1500	1.4342	1.3157
2_3^+		1.4310	1.2854	1.4964	1.4094

improve the description of nucleon scattering in the tungsten region at incident energies from 100 keV to 200 MeV.

In addition, this work investigates further the effect of volume conservation (a consequence of the assumption of the incompressibility of nuclear matter) in nuclear shape oscillations for the tungsten mass region. A static radius correction resulting from static axial-rigid deformations is introduced into the optical model radius. This radius correction does not lead to OMP coupling changes for any particular nucleus, but can be important for building regional potentials covering nuclei with varying deformations. The static radius correction allows removal of the mass number A dependence in OMP radii, therefore decreasing the number of OMP parameters.

The paper is structured as follows. Section 2 gives the nuclear model description, including model parametrization, OMP and potential radius correction. The results and discussions are presented in section 3. Finally, section 4 contains our conclusions.

Table 3. Coupling scheme for odd-*A* nuclides and corresponding core states. Energies of collective excited levels are available in ENSDF evaluated nuclear structure data [24, 25] (unit is MeV).

Core I^π	Nucleus I^π	^{183}W	Nucleus I^π	^{181}Ta
GS rotational band, first subband				
0^+	$(1/2)^-$	0.0000	$(7/2)^+$	0.0000
2^+	$(5/2)^-$	0.0991	$(11/2)^+$	0.3016
4^+	$(9/2)^-$	0.3089	$(15/2)^+$	0.7167
6^+	$(13/2)^-$	0.6311	$(19/2)^+$	1.2395
8^+	$(17/2)^-$	1.062	$(23/2)^+$	1.8631
10^+	$(21/2)^-$	1.5953	$(27/2)^+$	2.5801
GS rotational band, second subband				
0^+	$(3/2)^-$	0.0465	$(9/2)^+$	0.1363
2^+	$(7/2)^-$	0.2070	$(13/2)^+$	0.4952
4^+	$(11/2)^-$	0.4751	$(17/2)^+$	0.9650
6^+	$(15/2)^-$	0.8499	$(21/2)^+$	1.5393
8^+	$(19/2)^-$	1.3277	$(25/2)^+$	2.2101

2. Model description

2.1. Soft-rotator model parametrization

For all even–even nuclei (tungsten and hafnium isotopes) analyzed in this paper, we have fitted the level energies of their low-lying collective states using the soft-rotator model (SRM) [17, 18], with account of axial-rigid hexadecapole deformations and softness to symmetric quadrupole and octupole deformations with Hamiltonian parameters $\hbar\omega_0$, $\mu_{\beta_{20}}$, μ_{γ_0} , γ_0 , a_{32} , a_{42} , $\mu_{\epsilon}/\beta_{30}$, and δ_n . The experimental energies of these levels are well reproduced by the fitted SRM Hamiltonian with deviations lower than 10%. The resulting SRM Hamiltonian parameters and static deformation parameters of those levels are listed in table 1⁵. They are also used for calculation of the ‘effective’ deformations as discussed in references [14, 15].

The level schemes used for SRM fitting are listed in table 2. Note that, as shown in table 1, the level energies do not depend on deformation values explicitly, such as β_λ ($\lambda = 2, 4, \dots$), only in combination with other quantities; hence, SRM parameters do not contain these deformation values. They are determined only during the optical-model parameter fit. The coupling schemes used for optical calculations in this work are listed in tables 2 and 3. Levels of ^{182}W and ^{180}Hf were used as even–even cores of ^{183}W and ^{181}Ta , correspondingly, with the spin-level mapping shown in table 3. Multiple band coupling was considered for even–even nuclei [19–23]. However, only the GS band coupling was considered for odd-*A* nuclei since no assignment of bands based on the collective core excitations was found in the ENSDF database [24, 25] for ^{181}Ta and ^{183}W . For both even–even and odd-*A* nuclei, the calculation of the ‘effective’ deformations using SRM allowed us to account for nucleus stretching and dynamic volume conservation effects.

⁵ A detailed explanation of SRM Hamiltonian parameters can be found in references [14, 15].

2.2. Optical-model potential

A dispersive coupled-channels OMP for the analysis of scattering data is used in this work, which includes energy-dependent real V_i ($i = \text{HF}, v, s, C, \text{so}$) functionals for the Hartree–Fock (HF), volume (v), surface (s), Coulomb (C) and spin–orbit (so) potentials, respectively, and imaginary potentials W_i ($i = v, s, \text{so}$) of the same denomination. Imaginary potentials define the corresponding dispersive contributions ΔV_v (volume), ΔV_s (surface), and ΔV_{so} (spin–orbit). The general formulation of this potential has been published before (see e.g. reference [12]). However, since we will give the corresponding potential parameters, we summarize the OMP as follows:

$$\begin{aligned}
 V(r, R(\theta, \varphi), E) = & -V_{\text{HF}}(E^*) \times f_{\text{ws}}(r, R_{\text{HF}}(\theta, \varphi)) \\
 & - [\Delta V_v(E^*) + iW_v(E^*)] \times f_{\text{ws}}(r, R_v(\theta, \varphi)) \\
 & - [\Delta V_s(E^*) + iW_s(E^*)] \times g_{\text{ws}}(r, R_s(\theta, \varphi)) \\
 & + \left(\frac{\hbar}{m_\pi c} \right)^2 [V_{\text{so}}(E^*) + \Delta V_{\text{so}}(E^*) + iW_{\text{so}}(E^*)] \\
 & \times \frac{1}{r} \frac{d}{dr} f_{\text{ws}}(r, R_{\text{so}}) \times (\vec{\sigma} \cdot \vec{L}) \\
 & + V_{\text{Coul}}(r, R_c(\theta, \varphi)), \tag{1}
 \end{aligned}$$

where the ‘effective’ nucleon energy $E^* = E$ for neutrons and $E^* = E - E_{\text{Coul}}$ for protons. The estimate of the kinetic-energy loss of an incident proton in the Coulomb field is defined as $E_{\text{Coul}} = C_{\text{Coul}} \frac{Z'}{A^{1/3}}$, where Z' is the target atomic number and C_{Coul} is an undetermined constant.

The geometrical Woods–Saxon form factors are given as

$$\begin{aligned}
 f_{\text{ws}}(r, R_i(\theta, \varphi)) &= \{1 + \exp[(r - R_i(\theta, \varphi))/a_i]\}^{-1}, \quad i = \text{HF}, v \\
 g_{\text{ws}}(r, R_s(\theta, \varphi)) &= -4a_s \frac{d}{dr} f(r, R_s(\theta, \varphi)), \\
 f_{\text{ws}}(r, R_{0,\text{so}}) &= \{1 + \exp[(r - R_{0,\text{so}})/a_{\text{so}}]\}^{-1}, \\
 f_c(r, R_c(\theta, \varphi)) &= \{1 + \exp[(r - R_c(\theta, \varphi))/a_c]\}^{-1}, \tag{2}
 \end{aligned}$$

where the deformed radii $R_i(\theta, \varphi)$ are described by equation (10), and the radial mass number A dependence is taken as $R_i = r_i A^{\frac{1}{3}}$ for central potentials, with r_i being the reduced radii. A spherical spin–orbit potential is used with constant radius equal to $R_{0,\text{so}} = r_{0,\text{so}} A^{\frac{1}{3}}$. The deformed Coulomb potential $V_{\text{Coul}}(r, R_c(\theta, \varphi))$ is calculated using a multipole expansion of a charged ellipsoid with a uniform charge density within the Coulomb radius $R_c = r_c A^{\frac{1}{3}}$ and zero outside, as suggested by Bassel *et al* [26].

The energy dependence of the HF part of the nuclear mean field is taken as that found by Lipperheide [27, 28]:

$$V_{\text{HF}}(E) = A_{\text{HF}} \exp[-\lambda_{\text{HF}}(E - E_{\text{F}})], \tag{3}$$

where the parameters A_{HF} and λ_{HF} are undetermined constants and E_{F} is the Fermi energy. The energy dependencies for the imaginary volume $W_v(E)$ and surface $W_s(E)$ terms have been

Table 4. Dispersive coupled-channels OMP parameters for nucleon-induced reaction on tungsten and its adjacent nuclei.

	Volume	Surface	Spin-orbit	Coulomb
Real potential parameters	$V_0 = 51.65 + 0.00067$ (A-184) MeV $\lambda_{\text{HF}} = 0.00951 \text{ MeV}^{-1}$ $C_{\text{viso}} = 22.45 \text{ MeV}$		$V_{\text{so}} = 5.37 \text{ MeV}$ $\lambda_{\text{so}} = 0.005 \text{ MeV}^{-1}$	$C_{\text{Coul}} = 1 \text{ MeV}$
Imaginary potential parameters	$A_v = 12.29 \text{ MeV}$ $B_v = 77.03 \text{ MeV}$ $E_a = 54 \text{ MeV}$ $\alpha_v = 0.355 \text{ MeV}^{1/2}$	$W_0 = 14.08 \text{ MeV}$ $B_s = 11.03 \text{ MeV}$ $C_s = 0.01353 \text{ MeV}^{-1}$ $C_{\text{wiso}} = 22.2 \text{ MeV}$	$W_{\text{so}} = -3.1 \text{ MeV}$ $B_{\text{so}} = 160 \text{ MeV}$	
Potential geometry	$r_{\text{HF}} = 1.254 \text{ fm}$ $a_{\text{HF}} = 0.599 + 0.002$ (A-184) fm $r_v = 1.266 \text{ fm}$ $a_v = 0.626 - 0.0036$ (A-184) fm	$r_s = 1.164 \text{ fm}$ $a_s = 0.61 \text{ fm}$	$r_{\text{so}} = 1.121 \text{ fm}$ $a_{\text{so}} = 0.59 \text{ fm}$	$r_c = 1.123 \text{ fm}$ $a_c = 0.86 \text{ fm}$

suggested by Brown and Rho [29] and Delaroche *et al* [30], respectively, as follows:

$$W_v(E) = A_v \frac{(E - E_F)^2}{(E - E_F)^2 + (B_v)^2}, \quad (4)$$

$$W_s(E) = A_s \frac{(E - E_F)^2}{(E - E_F)^2 + (B_s)^2} \exp(-C_s |E - E_F|), \quad (5)$$

where A_v , B_v , A_s , B_s , and C_s are undetermined constants. These imaginary functionals are analytically integrable [31].

The dispersive OMP analysis of neutron scattering on ^{27}Al [32] showed the importance of using a nonsymmetric version of the volume imaginary potential for large positive and large negative energies to describe neutron total cross-section data above 100 MeV. Following Mahaux and Sartor [28], the assumption that the imaginary potential $W_v(E)$ is symmetric about $E = E_F$ [according to the equation $W(2E_F - E) = W(E)$] is modified above some fixed energy E_a , which is expected to be close to 60 MeV, but it is treated as a parameter. The nonsymmetric addition to $W_v(E)$ is proportional to the undetermined coefficient α_v as discussed in [12].

An OMP that contains an isovector term can be demonstrated to be approximately lane-consistent [33], and used to calculate nucleon scattering observables for nucleon-induced reactions including quasi-elastic (p, n) scattering. The isospin dependence of the current potential (the lane term [34, 35]) is considered in both real $V_{\text{HF}}(E)$ and imaginary surface $W_s(E)$ potentials as follows:

$$A_{\text{HF}} = V_0 \left[1 + (-1)^{(Z'+1)} \frac{C_{\text{viso}}}{V_0} \frac{(N - Z)}{A} \right], \quad (6)$$

$$A_s = W_0 \left[1 + (-1)^{(Z'+1)} \frac{C_{\text{wiso}}}{W_0} \frac{(N - Z)}{A} \right], \quad (7)$$

where V_0 , C_{viso} , W_0 and C_{wiso} are undetermined constants.

Table 5. GS deformation parameters of hafnium, tungsten and tantalum isotopes allowing best fit of experimental data versus default theoretical FRDM deformation parameters derived by Möller and Nix [44].

Target	β_2		β_4		β_6	
	Present	FRDM [44]	Present	FRDM [44]	Present	FRDM [44]
^{182}W	0.232	0.259	-0.064	-0.084	0.007	0.001
^{183}W	0.251	0.250	-0.060	-0.093	0.016	0.004
^{184}W	0.246	0.240	-0.074	-0.095	0.0119	0.010
^{186}W	0.235	0.230	-0.087	-0.107	0.0056	0.020
^{178}Hf	0.272	0.278	-0.048	-0.080	0.0054	-0.004
^{181}Ta	0.271	0.269	-0.072	-0.090	0.0028	0.001

The energy dependence of the spin-orbit potential is taken in the functional form suggested by Koning and Delaroche [36], but contains a dispersive contribution $\Delta V_{\text{so}}(E)$ [37]:

$$V_{\text{so}}(E) = V_{\text{so}} \exp[-\lambda_{\text{so}}(E - E_{\text{F}})], \quad (8)$$

$$W_{\text{so}}(E) = W_{\text{so}} \frac{(E - E_{\text{F}})^2}{(E - E_{\text{F}})^2 + (B_{\text{so}})^2}, \quad (9)$$

where V_{so} , λ_{so} , W_{so} , and B_{so} are undetermined constants.

The calculations of dispersive contributions are performed using dispersion relations, which are integrated as discussed in references [31, 38, 39]. Fermi energies E_{F} for the neutron and proton system are determined as $E_{\text{F}}(Z, A) = -\frac{1}{2}[S_n(Z, A) + S_n(Z, A + 1)]$ for neutrons and $E_{\text{F}}(Z, A) = -\frac{1}{2}[S_p(Z, A) + S_p(Z + 1, A + 1)]$ for protons, where $S_i(Z, A)$ denotes the corresponding separation energy of nucleon i from a nucleus labeled by Z and A .

The optical-model code OPTMAN [40, 41] was used for OMP parameter fitting and scattering data calculations. The OMP parameters were searched for by minimizing the quantity χ^2 [42]. All experimental data and related references used in the calculations were retrieved from the EXFOR database [43].

Table 4 lists the derived dispersive coupled-channel OMP parameters for the regional OMP. Table 5 contains best-fit static deformations for considered nuclides and their comparison with FRDM calculations [44], which are employed as the default deformation parameter set. Fitted deformations are in reasonable agreement with FRDM results.

2.3. Geometrical parameters of the deformed optical-model potential

2.3.1. OMP radii. The instant nuclear shape up to $\lambda = 8$ multipole deformations in a body-fixed system was described in reference [12] (see equations (1) and (2) there), as a function of dynamic variables β_2 , γ , β_3 , and η and static variables β_{40} , β_{60} , and β_{80} . If the β_2 deformation can be taken as a large static axial component β_{20} plus a small dynamic contribution $\delta\beta_2$ ⁶, then a truly small parameter $\delta\beta_2$ can be used for a Taylor expansion, instead of the relatively large parameter β_2 originally used by Tamura [13]. The departures from the axially symmetric rigid-rotor shape are considered to all orders in quadrupole nonaxiality parameter γ and octupole nonaxiality parameter η . Under these assumptions we can rewrite equation (1) of reference

⁶I.e. $\beta_2 = \beta_{20} + \delta\beta_2$, where $\beta_{20} \gg \delta\beta_2$, which is the usual case for actinides or rare-earth elements.

[12] as follows:

$$\begin{aligned}
 R_i(\theta', \varphi') = R_{0i} \left\{ \left(1 + \sum_{\lambda=2,4,6,8} \beta_{\lambda 0} Y_{\lambda 0}(\theta') \right) \right. \\
 + \beta_{20} \left[\frac{\delta\beta_2}{\beta_{20}} \cos \gamma + \cos \gamma - 1 \right] Y_{20}(\theta') \\
 + (\beta_{20} + \delta\beta_2) \frac{\sin \gamma}{\sqrt{2}} [Y_{22}(\theta', \varphi') + Y_{2-2}(\theta', \varphi')] \\
 \left. + \beta_3 \cos \eta Y_{30}(\theta') + \beta_3 \frac{\sin \eta}{\sqrt{2}} [Y_{32}(\theta', \varphi') + Y_{3-2}(\theta', \varphi')] \right\}. \quad (10)
 \end{aligned}$$

Equation (10) can be rewritten by explicitly separating the static axially symmetric (rigid) and dynamic (soft) terms:

$$\begin{aligned}
 R_i(\theta', \varphi') = R_{0i} \left\{ 1 + \sum_{\lambda=2,4,6,8} \beta_{\lambda 0} Y_{\lambda 0}(\theta') + f_{\text{soft}}(\delta\beta_2, \beta_3, \gamma, \eta; \theta', \varphi') \right\} \\
 = R_{0i} \{ f_{\text{rigid}}(\beta_{20}, \beta_{40}, \beta_{60}, \beta_{80}; \theta') + f_{\text{soft}}(\delta\beta_2, \beta_3, \gamma, \eta; \theta', \varphi') \}, \quad (11)
 \end{aligned}$$

where $R_{0i} f_{\text{rigid}}(\beta_{20}, \beta_{40}, \beta_{60}, \beta_{80}; \theta')$ describes an axial rigid rotor nuclear potential shape, and $R_{0i} f_{\text{soft}}(\delta\beta_2, \beta_3, \gamma, \eta; \theta', \varphi')$ describes the dynamical departure from the rigid axial shape due to quadrupolar and octupolar axial and nonaxial vibrations.

For a deformed nucleus with the shape of equation (10), its shape oscillates around rigid-rotor deformations and thus its volume changes as a function of the deformation. These volume changes can be compensated by adding a negative dynamical zero monopole β_{00} deformation into the optical model radius as done in reference [45], and therefore guaranteeing the volume conservation in the nuclear structure model [18] as follows:

$$\begin{aligned}
 R_i(\theta', \varphi') = R_{0i} \{ f_{\text{rigid}}(\beta_{20}, \beta_{40}, \beta_{60}, \beta_{80}; \theta') \\
 + \beta_{00} Y_{00} + f_{\text{soft}}(\delta\beta_2, \beta_3, \gamma, \eta; \theta', \varphi') \}. \quad (12)
 \end{aligned}$$

Such additional monopole deformation term β_{00} implies an additional zero-potential multipole allowing coupling with non-GS bands [15, 46].

The shape given by equation (12) leads to a deformable nuclear volume V_{def} . If we take up to the first nonzero order in each $\beta_{\lambda\mu}$ considered, we can write

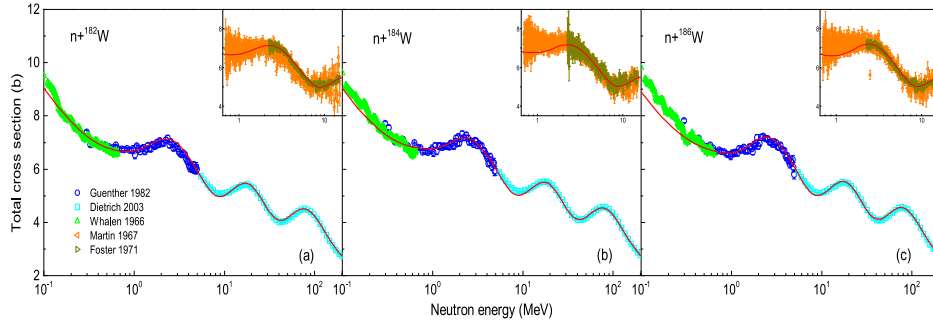
$$V_{\text{def}} = \frac{4\pi}{3} R_{0i}^3 \left[1 + \frac{3}{4\pi} \beta_{00} + \frac{3}{4\pi} \left(\sum_{\lambda=2,4,6,8} \beta_{\lambda 0}^2 + 2\beta_{20}\delta\beta_2 + \delta\beta_2^2 + \beta_3^2 \right) \right]. \quad (13)$$

The monopole deformation β_{00} can be deduced from equation (13) by equalizing the deformable nuclear volume V_{def} to the volume at zero deformation $\frac{4\pi}{3} R_{0i}^3$, so we obtain

$$\beta_{00} = -\frac{1}{\sqrt[4]{\pi}} \left(\sum_{\lambda=2,4,6,8} \beta_{\lambda 0}^2 + 2\beta_{20}\delta\beta_2 + \delta\beta_2^2 + \beta_3^2 \right) \equiv \beta_{00}^{\text{rig}} + \beta_{00}^{\text{dyn}} \quad (14)$$

Table 6. Comparison of calculated and experimental average resonance parameters. The values inside parentheses indicate uncertainties.

Quantity	Reference	^{182}W	^{183}W	^{184}W	^{186}W	^{178}Hf	^{181}Ta
S_0 [(eV) $^{-1/2}$ 10^{-4}]	Present	2.22	1.95	2.14	2.12	1.99	1.63
	RIPL-3 [10]	2.3(0.3)	1.9(0.3)	2.5(0.4)	2.3(0.4)	2.1(0.3)	1.7(0.2)
	Mughabghab 2006 [49]	2.51(0.3)	1.89(0.29)	2.52(0.32)	2.22(0.32)	2.2(0.7)	1.74(0.12)
S_1 [(eV) $^{-1/2}$ 10^{-4}]	Present	0.71	0.76	0.72	0.77	0.88	0.77
	Mughabghab 2006 [49]	0.72(0.06)	0.75(0.03)	0.72(0.07)	0.72(0.2)	0.80(0.14)	0.5(0.2)
R' (fm)	Present	7.86	7.72	7.83	7.85	8.05	7.61
	Mughabghab 2006 [49]	7.67(0.06)	8.1(0.1)	8.1(0.04)	8.52(0.02)	7.5(0.3)	7.65(0.2)

**Figure 1.** Comparison of total cross sections for $^{182,184,186}\text{W}$ with experimental data [4, 9, 50–52] from 100 keV to 200 MeV. The smaller plot shows the comparison with those data by Martin [51] and Foster [52].

where the first term,

$$\beta_{00}^{\text{rig}} \equiv -\frac{1}{4\pi} \sum_{\lambda=2,4,6,8} \beta_{\lambda 0}^2, \quad (15)$$

is the static volume correction to the radius, which depends only on static axial-rigid deformations.

The second term,

$$\beta_{00}^{\text{dyn}} = -\frac{1}{4\pi} (2\beta_{20}\delta\beta_2 + \delta\beta_2^2 + \beta_3^2), \quad (16)$$

is a dynamic correction to radius, which leads to an additional coupling of levels with the same spin from different bands in coupled-channels equations.

In the previous publication [15] only the dynamic monopole correction β_{00}^{dyn} to the radius was considered. Such a correction should be applied to all nuclei used in this work. The dynamic volume conservation does affect the predictions of the optical observables, most significantly for incident particle energies up to several MeV. At higher energies, a small decrease in calculated cross sections can be attributed to the decrease in effective nuclear radius.

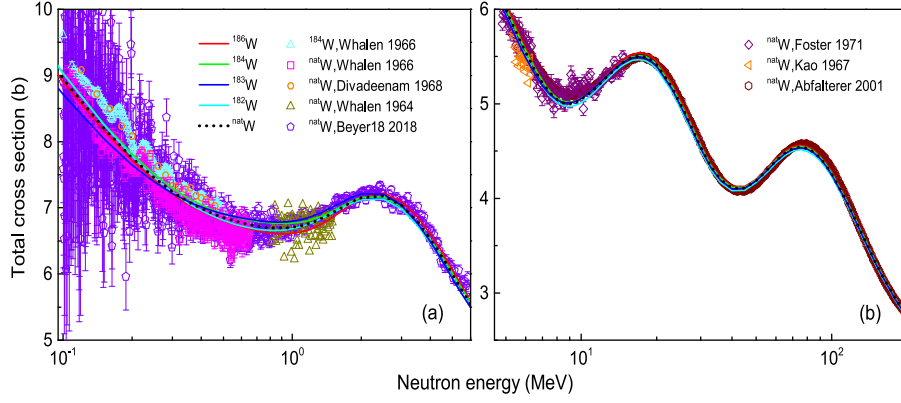


Figure 2. Comparison of total cross sections of natural tungsten with experimental data [50, 52–57]. The total cross sections for individual isotopes are also shown. The left panel shows the results below 6 MeV down to 0.1 MeV, while the right panel is the result above 4.5 MeV up to 200 MeV.

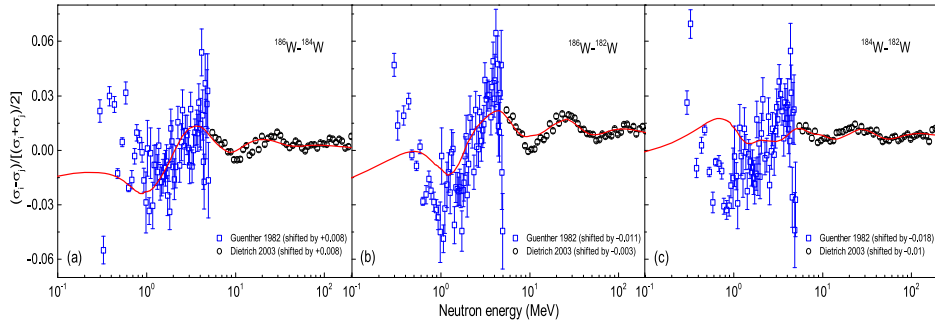


Figure 3. Ratio of total cross-section difference for different tungsten isotopes from 0.1 to 200 MeV, compared with experimental data [4, 9].

However, if we intend to deduce a systematic of regional optical potential parameters for nuclei in the selected Z, A range, then the static deformation correction β_{00}^{rig} should also be considered since all nuclei have different nuclear densities due to different static deformations. The β_{00}^{rig} correction is significant if we intend to keep constant constant nuclear volume for nuclei with different deformations.

Since $\beta_{00}^{\text{rig}} Y_{00}$ is a constant parameter for a certain nuclide, we can include it in the potential radius $R_{0,i}$ in equation (12); $\beta_{00}^{\text{dyn}} Y_{00}$ should be included in the functional $f_{\text{soft}}(\delta\beta_2, \beta_3, \gamma, \eta; \theta', \varphi')$ as in reference [15], so we obtain

$$R_i(\theta', \varphi') = \tilde{R}_{0i}(\beta_{\lambda 0}) \{ f_{\text{rigid}}(\beta_{20}, \beta_{40}, \beta_{60}, \beta_{80}; \theta') + \tilde{f}_{\text{soft}}(\delta\beta_2, \beta_3, \gamma, \eta; \theta', \varphi') \}, \quad (17)$$

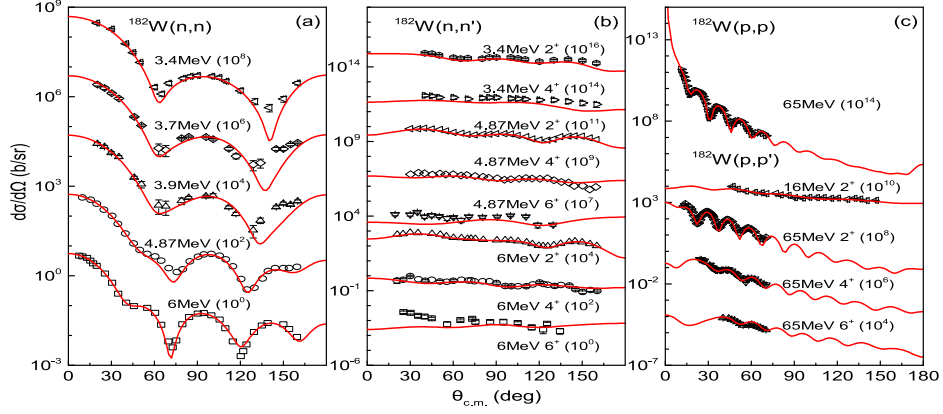


Figure 4. Nucleon elastic and inelastic scattering angular distributions for ^{182}W at different incident energies, compared with the experimental data [4, 7, 62–65].

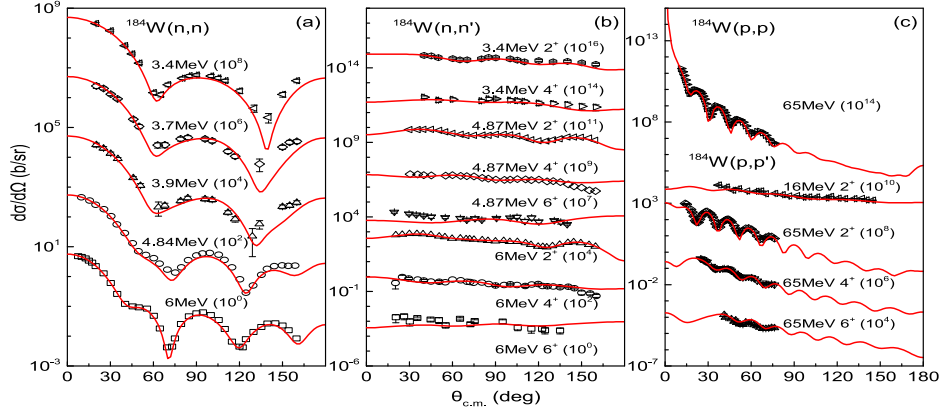


Figure 5. Same as figure 4, but for ^{184}W targets, compared with the experimental data [4, 7, 62–65].

where

$$\tilde{R}_{0i}(\beta_{\lambda 0}) \equiv R_{0i} \left(1 + \beta_{00}^{\text{rig}} Y_{00} \right) = r_{0i} A^{1/3} \left(1 - \frac{1}{\sqrt[4]{\pi}} \sum_{\lambda=2,4,6,8} \beta_{\lambda 0}^2 \right), \quad (18)$$

r_{0i} is a corresponding reduced radius, and

$$\tilde{f}_{\text{soft}}(\delta\beta_2, \beta_3, \gamma, \eta; \theta', \varphi') \equiv \beta_{00}^{\text{dyn}} Y_{00} + f_{\text{soft}}(\delta\beta_2, \beta_3, \gamma, \eta; \theta', \varphi'). \quad (19)$$

We can see that now the nuclear radius given by equation (18) depends explicitly not only on the nucleus mass A but also on the static deformation $\beta_{\lambda 0}$ parameters for individual nuclides. Note that we keep the form of the nuclear shape in equation (10) upon introduction of the volume conservation through the new radius $\tilde{R}_{0i}(\beta_{\lambda 0})$ (depending on axial-rigid deformations) and the monopolar term into the dynamic factor f_{soft} . This splitting helps to incorporate the volume conservation effects into existing calculation routines. All nuclear radii of the optical potential

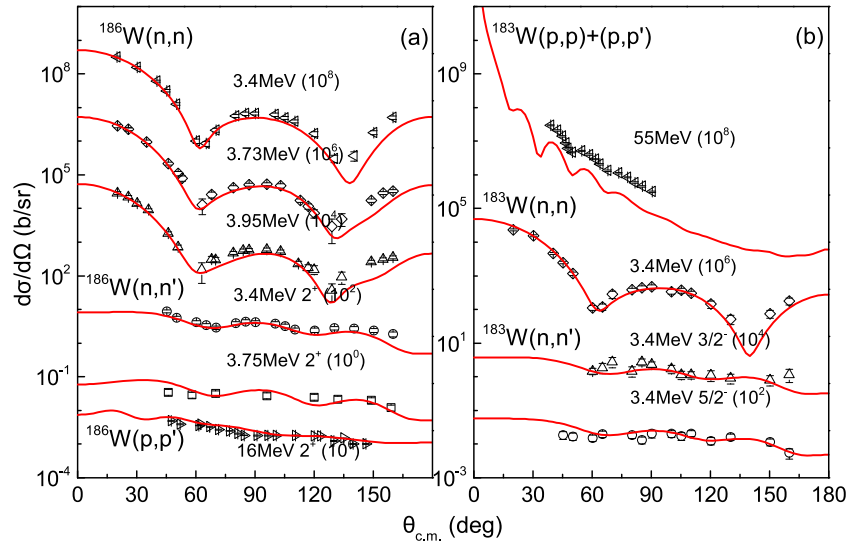


Figure 6. Same as figure 4, but for ^{186}W and ^{183}W , compared with the experimental data [4, 7, 63, 66]. The label ‘(p, p)+(p, p)’ implies the sum of elastic and inelastic scattering angular distributions of the first, second, third, fourth and fifth excited states.

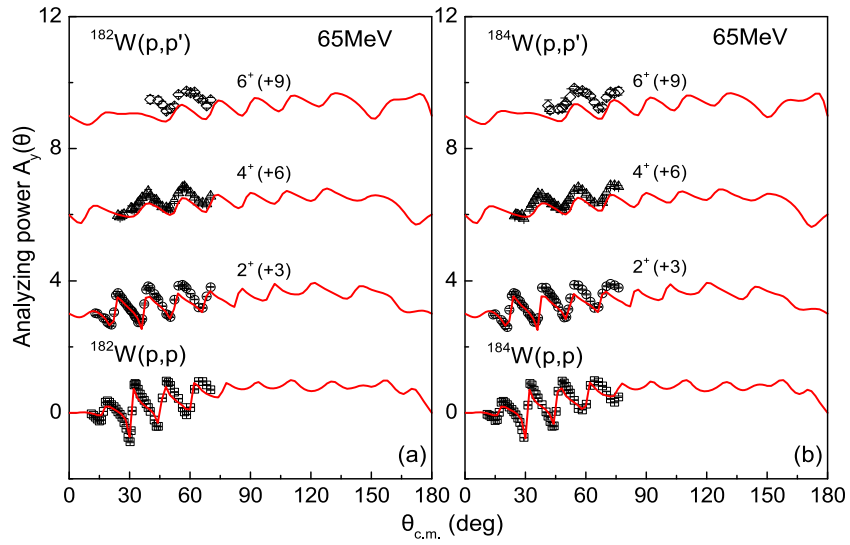


Figure 7. Proton elastic (on bottom) and inelastic scattering analyzing powers A_y for tungsten isotopes, compared with the experimental data [64, 65].

are changed according to this recipe. This allows removal of the mass number A dependence on radii, therefore decreasing the number of free optical potential parameters as can be seen in table 4.

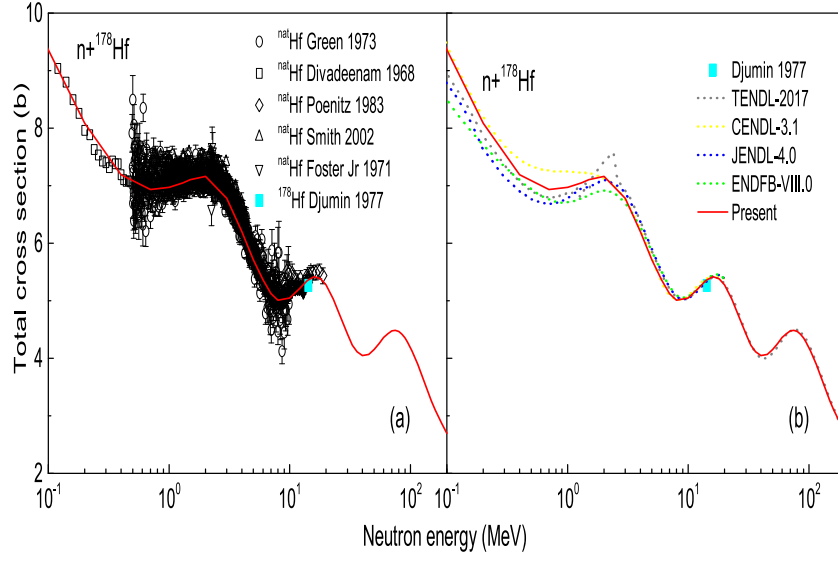


Figure 8. Total cross sections for ^{178}Hf . The left panel shows the results for ^{178}Hf , compared with experimental data of ^{178}Hf and $^{\text{nat}}\text{Hf}$ [52, 53, 67–70]; the right panel shows the results for ^{178}Hf , compared with estimations from libraries from 100 keV to 200 MeV.

2.3.2. OMP diffuseness. All above considerations concern an OMP that follows the nuclear density with a sharp edge. For medium- and heavy-mass nuclei a Woods–Saxon form represents a much better approximation of the nuclear density that features a finite diffuse surface [47]:

$$\rho(r) = \frac{\rho_0}{1 + e^{(r-R)/a}},$$

where R is the half-maximum radius and a is the diffuseness. Then, in order to conserve the nuclear mass as shown in [46], the zero multipole β_{00} is multiplied by a correction coefficient C^{diff} : $\beta'_{00} = C^{\text{diff}}\beta_{00}$.

Using the approximate formula suggested in reference [48], the correction coefficient C^{diff} can be evaluated analytically:

$$C^{\text{diff}} = \frac{1}{1 + \frac{1}{3}\left(\frac{\pi a}{R}\right)^2}.$$

C^{diff} increases with the square of the nuclear radius, i.e. with $A^{\frac{2}{3}}$. It is approximately 0.85 for ^{12}C [46], but it is about 0.97 for the nuclei considered in this work. This is physically clear as the ratio of the nuclear boundary volume to the full nuclear volume quickly decreases as the radius grows. Therefore, this correction can be neglected in this work.

3. Results and discussion

This section presents the calculations of cross sections, angular distributions and analyzing power for $^{182,184,186}\text{W}$ isotopes and natural tungsten.

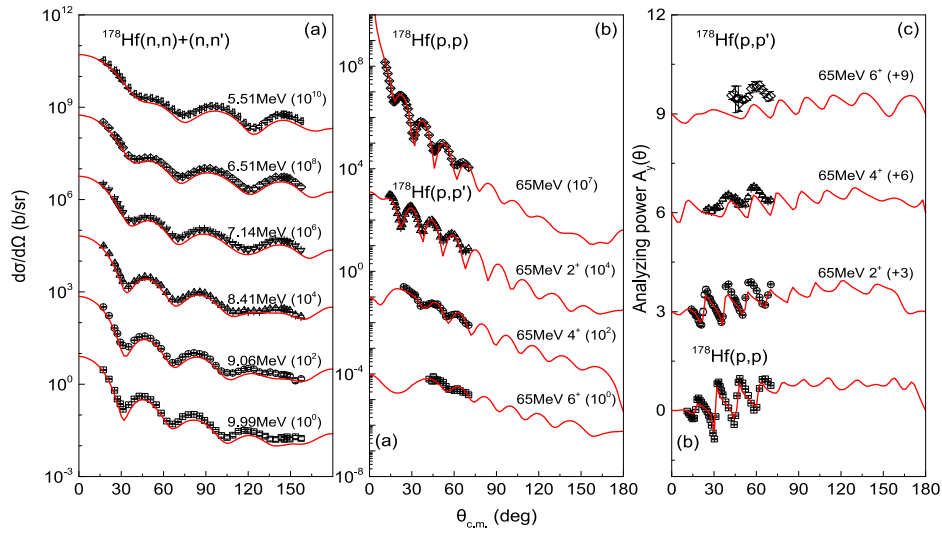


Figure 9. Nucleon scattering angular distributions and analyzing powers for ^{178}Hf , compared with the experimental data [64]. The left panel shows the results of neutron scattering angular distributions, compared with experimental data of $^{\text{nat}}\text{Hf}$ [67]. The label ‘(n, n) + (n, n)’ implies the sum of elastic and inelastic scattering angular distributions of the first and second excited states.

Table 6 shows the calculated average resonance parameters compared with RIPL-3 [10] and Mughabghab [49] evaluations. It can be seen that these data are reasonably described within evaluated uncertainties by the present optical model, except for the case of S_1 for ^{181}Ta .

Figure 1 shows the calculated neutron total cross sections for $^{182,184,186}\text{W}$ isotopes. Calculations are in excellent agreement with experimental data from 0.1 to 200 MeV. Note a slight underestimation of Whalen 1965 data at low energies, especially for the ^{186}W target.

Figure 2 shows a comparison of measured and calculated neutron total cross sections⁷ on a natural tungsten target. The left panel shows the comparison below 6 MeV. Reasonable agreement with Whalen [50] and Beyer data [55] is shown in the whole energy range, while the Divadeenam [53] data are higher than calculations below 300 keV. Excellent agreement of the calculated neutron total cross sections with measurements is achieved with Foster [52] and Abfalterer [57] data as shown in the right panel of figure 2.

Figure 3 presents a ratio of total cross-section differences for different pairs of tungsten isotopes from 0.1 MeV to 200 MeV. The ratio $R_{i,j}$ of isotopes i and j is defined as the cross-section difference over the average cross section, i.e., $R_{i,j} = (\sigma_i - \sigma_j) / ((\sigma_i + \sigma_j)/2)$. This quantity is very well measured and highly sensitive to the quality of derived optical potential as discussed in references [58–60].

Dietrich and colleagues measured neutron total cross-section differences among the isotopes $^{182,184,186}\text{W}$ in the 5–560 MeV energy range [9]. Their measurements showed oscillations up to 150 mb between 5 and 100 MeV, which could not be described in their publication by phenomenological potentials [9]. Dietrich *et al* suggested that *only a microscopic potential that includes a realistic description of the neutron and proton density distributions is capable*

⁷ Reconstructed from the sum of abundance-weighted total cross sections of four isotopes $^{182,183,184,186}\text{W}$; the contribution from ^{180}W is neglected since its abundance is only 0.13%.

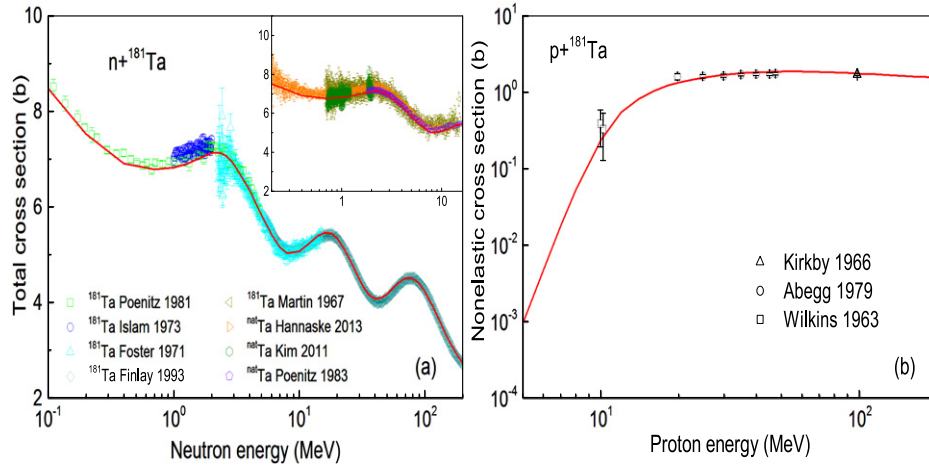


Figure 10. Neutron total cross sections and proton nonelastic cross section for ^{181}Ta . The left panel shows the results of neutron total cross sections, compared with experimental data [51, 52, 69, 71–75] from 100 keV to 200 MeV (the smaller plot shows the comparison with those data for natural tantalum by [51, 69, 74, 75]); the right panel shows the results of the proton nonelastic cross section, compared with experimental data [76–79].

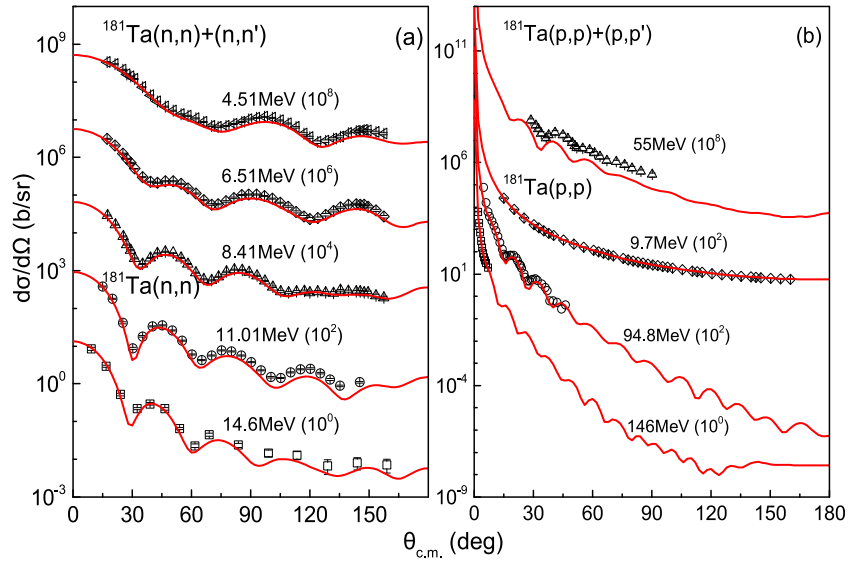


Figure 11. Nucleon scattering angular distributions for ^{181}Ta , compared with the experimental data [66, 80–85]. The label ‘(n, n) + (n, n)’ implies the sum of elastic and inelastic scattering angular distributions of the first, second and third excited states; the label ‘(p, p) + (p, p)’ implies the sum of elastic and inelastic scattering angular distributions of the first, second, third, fourth and fifth excited states.

of describing their data [9]. However, dispersive coupled-channel phenomenological potentials that include isovector terms and use accurate nuclear structure models for couplings have

proven capable of describing measured total cross-section differences for actinides [61]. The deformed dispersive optical model featuring an isovector term derived in this work provides an excellent description of Dietrich data as can be seen in figure 3.

Figure 4 presents the nucleon elastic and inelastic scattering angular distributions for ^{182}W , while figures 5 and 6 show the results for ^{184}W , ^{186}W and ^{183}W , respectively. The nucleon scattering cross sections for these isotopes are described fairly well, with only slight underestimation at low energies less than 5 MeV, probably due to missing compound-nucleus contributions. For the case of ^{183}W , the prediction of proton elastic scattering data at 55 MeV underestimates the experimental data. Perhaps experimental issues affected those data, which were the only ones of that work to be acquired with oxide targets.

Figure 7 shows the proton elastic/inelastic scattering analyzing powers A_y for tungsten isotopes. It can be seen that the predictions for elastic scattering A_y are excellent. Those for inelastic scattering A_y are also rather good. However, the predictions for inelastic scattering A_y are of somewhat lower quality than those for elastic scattering A_y . Meanwhile, slight underestimation near the extrema can be seen for the cases of the 2^+ , 4^+ , and 6^+ states.

In addition, we present some calculations compared with data for neighboring nuclei ^{178}Hf and ^{181}Ta in figures 8–11. A similar agreement is observed to the one discussed above for tungsten isotopes. ^{181}Ta results also show the excellent description achieved for odd- A nuclei.

4. Conclusions

This work presents a nucleon scattering analysis using a newly fitted lane-consistent dispersive optical potential for hafnium, tungsten and tantalum isotopes for incident nucleons with energies from 100 keV up to 200 MeV. The OMP couplings are based on an SRM of even–even target nuclei, and an approximated model to describe odd targets. Fitted static deformations are in reasonable agreement with FRDM calculations [44]. Excellent agreement is obtained with available experimental nucleon scattering data. Our model allows good description of total cross-section ratios together with other nucleon scattering data by a single dispersive potential with full isovector terms, contrary to the conclusions from reference [9].

The implemented radius correction procedure based on nuclear volume conservation allowed one to remove the A -dependence of the radii of the derived optical potentials. Consequently, it decreased the number of free parameters.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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References

- [1] Bolt H, Barabash V, Krauss W, Linke J, Neu R, Suzuki S and Yoshida N (ASDEX Upgrade Team) 2004 Materials for the plasma-facing components of fusion reactors *J. Nucl. Mater.* **329–33** 66
- [2] Neu R *et al* 2007 Final steps to an all tungsten divertor tokamak *J. Nucl. Mater.* **363–5** 52
- [3] Zhang S *et al* 2015 Measurement of leakage neutron spectra for tungsten with D-T neutrons and validation of evaluated nuclear data *Fusion Eng. Des.* **92** 41
- [4] Guenther P T, Smith A B and Whalen J F 1982 Fast-neutron total and scattering cross sections of ^{182}W , ^{184}W , and ^{186}W *Phys. Rev. C* **26** 2433
- [5] Jacobsen S B 2005 The Hf–W isotopic system and the origin of the earth and moon *Annu. Rev. Earth Planet. Sci.* **33** 531
- [6] Kruijer T S and Kleine T 2017 Tungsten isotopes and the origin of the moon *Earth Planet. Sci. Lett.* **475** 15
- [7] Delaroche J P, Haouat G, Lachkar J, Patin Y, Sigaud J and Chardine J 1981 Deformations, moments, and radii of $^{182,183,184,186}\text{W}$ from fast neutron scattering *Phys. Rev. C* **23** 136
- [8] Romain P and Delaroche J-P 1996 *Proc. of a Specialists' Meeting on the Nucleon–Nucleus Optical Model up to 200 MeV, Bruyères-le-Châtel* (France) (OECD, Paris, France, 1997) p 167
- [9] Dietrich F S *et al* 2003 Importance of isovector effects in reproducing neutron total cross section differences in the W isotopes *Phys. Rev. C* **67** 044606
- [10] Capote R *et al* 2009 RIPL-reference input parameter library for calculation of nuclear reactions and nuclear data evaluations *Nucl. Data Sheets* **110** 3107–214
- [11] Dietrich F S, Thompson I J and Kawano T 2012 Target-state dependence of cross sections for reactions on statically deformed nuclei *Phys. Rev. C* **85** 044611
- [12] Soukhovitskiĭ E S, Capote R, Quesada J M, Chiba S and Martyanov D S 2016 Nucleon scattering on actinides using a dispersive optical model with extended couplings *Phys. Rev. C* **94** 064605
- [13] Tamura T 1965 Analyses of the scattering of nuclear particles by collective nuclei in terms of the coupled-channel calculation *Rev. Mod. Phys.* **37** 679
- [14] Martyanov D S, Soukhovitskiĭ E S, Capote R, Quesada J M and Chiba S 2017 Predicting the optical observables for nucleon scattering on even-even actinides *Chin. Phys. C* **41** 094105
- [15] Martyanov D, Soukhovitskiĭ E, Capote R, Quesada J M and Chiba S 2017 Optical model with multiple band couplings using soft rotator structure *EPJ Web Conf.* **146** 12031
- [16] Martyanov D S, Soukhovitskiĭ E S, Capote R, Quesada J M and Chiba S 2020 Multiband coupling and nuclear softness in optical model calculations for even-even and odd-A actinides *EPJ Web of Conferences* **239** 03003
- [17] Porodzinskiĭ Y V and Soukhovitskiĭ E Sh 1996 Rotational-vibrational states of nonaxial deformable even-even nuclei *Yad. Fiz.* **53** 64 64
- [18] Porodzinskiĭ Y V and Soukhovitskiĭ E S 1991 *Sov. J. Nucl. Phys.* **53** 41
- [18] Porodzinskiĭ Y V and Sukhovitskiĭ E S 1996 Analysis of neutron scattering by even–even nuclei with allowance for dynamical octupole deformations *Yad. Fiz.* **59** 247
- [18] Porodzinskiĭ Y V and Sukhovitskiĭ E S 1996 *Phys. Atom. Nucl.* **59** 228
- [19] Singh B 2015 Nuclear data sheets for $A = 182$ *Nucl. Data Sheets* **130** 21
- [20] Baglin C M 2010 Nuclear data sheets for $A = 184$ *Nucl. Data Sheets* **111** 275
- [21] Baglin C M 2003 Nuclear data sheets for $A = 186$ *Nucl. Data Sheets* **99** 1
- [22] Achterberg E, Capurro O A and Marti G V 2009 Nuclear data sheets for $A = 178$ *Nucl. Data Sheets* **110** 1473
- [23] McCutchan E A 2015 Nuclear data sheets for $A = 180$ *Nucl. Data Sheets* **126** 151
- [24] Baglin C M 2016 Nuclear data sheets for $A = 183$ *Nucl. Data Sheets* **134** 149
- [25] Wu S-C 2005 Nuclear data sheets for $A = 181$ *Nucl. Data Sheets* **106** 367
- [26] Bassel R H, Drisko R M and Satchler G R 1962 The distorted wave theory of direct nuclear reactions *Technical Report ORNL-3240* Oak Ridge National Laboratory Oak Ridge, TN
- [27] Lipperheide R 1966 On the nuclear optical-model potential *Nucl. Phys.* **89** 97
- [28] Mahaux C and Sartor R 1991 Dispersion relation approach to the mean field and spectral functions of nucleons in ^{40}Ca *Nucl. Phys. A* **528** 253
- [29] Brown G E and Rho M 1981 The giant Gamow–Teller resonance *Nucl. Phys. A* **372** 397
- [30] Delaroche J P, Wang Y and Rapaport J 1989 Neutron- ^{90}Zr mean field from a dispersive optical model analysis *Phys. Rev. C* **39** 391
- [31] Quesada J M, Capote R, Molina A, Lozano M and Raynal J 2003 Analytical expressions for the dispersive contributions to the nucleon–nucleus optical potential *Phys. Rev. C* **67** 067601

- [32] Molina A, Capote R, Quesada J M and Lozano M 2002 Dispersive spherical optical model of neutron scattering from 27 Al up to 250 MeV *Phys. Rev. C* **65** 034616
- [33] Quesada J M, Capote R, Soukhovitskiĭ E S and Chiba S 2007 Approximate lane consistency of the dispersive coupled-channels potential for actinides *Phys. Rev. C* **76** 057602
- [34] Lane A M 1962 New term in the nuclear optical potential: implications for (p, n) mirror state reactions *Phys. Rev. Lett.* **8** 171
- [35] Lane A M 1962 Isobaric spin dependence of the optical potential and quasi-elastic (p, n) reactions *Nucl. Phys.* **35** 676
- [36] Koning A J and Delaroche J P 2003 Local and global nucleon optical models from 1 keV to 200 MeV *Nucl. Phys. A* **713** 231
- [37] Morillon B and Romain P 2004 Dispersive and global spherical optical model with a local energy approximation for the scattering of neutrons by nuclei from 1 keV to 200 MeV *Phys. Rev. C* **70** 014601
- [38] Capote R, Molina A and Quesada J M 2001 A general numerical solution of dispersion relations for the nuclear optical model *J. Phys. G: Nucl. Part. Phys.* **27** B15–9
- [39] Quesada J M, Capote R, Molina A and Lozano M 2003 Dispersion relations in the nuclear optical model *Comput. Phys. Commun.* **153** 97–105
- [40] Soukhovitskiĭ E S, Morogovskii G B, Chiba S, Iwamoto O and Fukahori T 2004 Physics and numerical methods of OPTMAN: a coupled-channels method based on soft-rotator model for a description of collective nuclear structure and excitation *Technical Report JAERI-Data/Code 2004-002* Japan Atomic Energy Research Institute
- [41] Soukhovitskiĭ E S, Chiba S, Iwamoto O, Shibata K, Fukahori T and Morogovskii G B 2005 Programs OPTMAN and SHEMMAN version 8 *Technical Report JAERI-Data/Code 2005-002* Japan Atomic Energy Research Institute
- [42] Soukhovitskiĭ E S, Lee Y-O, Chang J, Chiba S and Iwamoto O 2000 Nucleon interaction with ^{58}Ni up to 150 MeV studied in the coupled-channels approach based on the soft-rotator nuclear structure model *Phys. Rev. C* **62** 044605
- [43] EXchange FORmat database (EXFOR) is maintained by the Network of Nuclear Reaction Data Centers (www.nds.iaea.org/nrdc/) (www.nds.iaea.org/exfor/)
- [44] Möller PöNix J R, Myers W D and Swiatecki W J 1995 Nuclear ground-state masses and deformations *At. Data Nucl. Data Tables* **59** 185
- [45] Eisenberg J M and Greiner W 1987 *Nuclear Theory: Nuclear Models* (Amsterdam: North-Holland)
- [46] Soukhovitskiĭ E S, Chiba S and Iwamoto O 1999 Coupled-channels optical model calculations with account of nuclear volume conservation *Nucl. Phys. A* **646** 19–28
- [47] Woods R D and Saxon D S 1954 Diffuse surface optical model for nucleon–nuclei scattering *Phys. Rev.* **95** 577
- [48] Elton L R B 1958 A semi-empirical formula for the nuclear radius *Nucl. Phys.* **5** 173
- [49] Mughabghab S F 2006 *Atlas of Neutron Resonances* 5th edn (Amsterdam: Elsevier)
- [50] Whalen J F and Meadows J W 1966 Fast neutron total cross sections using a mono-energetic source and an automated facility *Report ANL-7210* Argonne National Laboratory
- [51] Martin R C *et al* 1967 MeV neutron total cross sections of Ta and W isotopes *Bull. Am. Phys. Soc.* **12** 106
- [52] Foster D G and Glasgow D W 1971 Neutron total cross sections, 2.5–15 MeV. I. Experimental *Phys. Rev. C* **3** 576
- [53] Divadeenam M, Bilpuch E G and Newson H W 1968 Strength functions and the optical model *Diss. Abstr. Int.* **28-09** 3834
- [54] Whalen J F, Roge R and Smith A B 1964 Fast-neutron total cross sections with an automated facility *Bull. Am. Phys. Soc.* **9** 651
- [55] Beyer R *et al* 2018 The neutron transmission of ^{197}Au , ^{nat}Fe and ^{nat}W *Eur. Phys. J. A* **54** 81
- [56] Kao W H *et al* 1967 Total neutron cross sections of sulfur and tungsten between 5.3 and 6.3 MeV *Chin. J. Phys.* **5** 43
- [57] Abfalterer W P, Bateman F B, Dietrich F S, Finlay R W, Haight R C and Morgan G L 2001 Measurement of neutron total cross sections up to 560 MeV *Phys. Rev. C* **63** 044608
- [58] Shamu R E, Bernstein E M, Ramirez J J and Lagrange C 1980 Effects of deformation on neutron total cross sections of even-A Nd and Sm isotopes *Phys. Rev. C* **22** 1857
- [59] Camarda H S, Phillips T W and White R M 1984 Neutron absolute and total cross section difference measurements in the mass-140 region *Phys. Rev. C* **29** 2106

- [60] Camarda H S, Dietrich F S and Phillips T W 1989 Microscopic optical-model calculations of neutron total cross sections and cross section differences *Phys. Rev. C* **39** 1725
- [61] Capote R, Soukhovitskii E S, Soukhovitskii E S, Quesada J M and Chiba S 2005 Is a global coupled-channel dispersive optical model potential for actinides feasible? *Phys. Rev. C* **72** 064610
- [62] Annand J R M and Finlay R W 1985 Direct neutron scattering from ^{182}W and ^{184}W *Nucl. Phys. A* **442** 234
- [63] Kruse T, Makofske W, Ogata H, Savin W, Slagowitz M, Williams M and Stoler P 1971 Elastic and inelastic proton scattering from heavy collective nuclei *Nucl. Phys. A* **169** 177
- [64] Ogawa H *et al* 1986 Inelastic scattering of 65-MeV polarized protons from ^{178}Hf , ^{180}Hf , ^{182}W , and ^{184}W and multipole moments of the optical potential *Phys. Rev. C* **33** 834
- [65] Ichihara T, Sakaguchi H, Nakamura M, Yosoi M, Ieiri M, Takeuchi Y, Togawa H, Tsutsumi T and Kobayashi S 1987 Inelastic proton scattering exciting the γ -vibrational band in deformed nuclei ($152 \leq A \leq 192$) at 65 MeV and the systematics of the hexadecapole (Y_{42}) strength of the γ vibration *Phys. Rev. C* **36** 1754
- [66] Kamitsubo H, Ohnuma H, Ôno K, Uchida A, Imaizumi M, Kobayashi S and Sekiguchi M 1967 Elastic scattering of 55 MeV protons from heavy nuclei *J. Phys. Soc. Japan* **22** 19
- [67] Smith A B 2002 The interaction of fast neutrons with hafnium *Ann. Nucl. Energy* **29** 1241
- [68] Green L and Mitchell J A 1973 Total cross section measurements with a ^{252}Cf time-of-flight spectrometer (LWBR development program) (United States)
- [69] Poenitz W P and Whalen J F 1983 Neutron total cross section measurements in the energy region from 47 keV to 20 MeV *Report ANL-NDM-80* (Argonne National Laboratory)
- [70] Djumin A N *et al* 1977 Nuclear radii for some isotopes derived from total from total neutron cross-sections *All Union Conf. on Neutron Physics* (Kiev) 18 C22
- [71] Poenitz W P, Whalen J F and Smith A B 1981 Total neutron cross sections of heavy nuclei *Nucl. Sci. Eng.* **78** 333
- [72] Islam E, Hussain M, Ameen N, Enayetullah M and Islam N 1973 Total cross sections of Y, Zr, La, Ce, Pr, Gd, Ho, Ta and Hg for 1.0 to 2.0 MeV neutrons *Nucl. Phys. A* **209** 189
- [73] Finlay R W, Abfalterer W P, Fink G, Montei E, Adami T, Lisowski P W, Morgan G L and Haight R C 1993 Neutron total cross sections at intermediate energies *Phys. Rev. C* **47** 237
- [74] Hannaske R *et al* 2013 Neutron total cross section measurements of gold and tantalum at the nELBE photoneutron source *Eur. Phys. J. A* **49** 137
- [75] Kim G D, Woo H J, Yang T K, Lee S Y and Lee Y O 2011 Measurement of fast neutron total cross sections on ^{nat}Ta and ^{nat}Bi in the MeV energy range *J. Korean Phy. Soc.* **59** 2233
- [76] Wilkins B D and Igo G 1963 10-MeV proton reaction cross sections for several elements *Phys. Rev.* **129** 2198
- [77] Kirkby P and Link W T 1966 Faraday-cup measurement of proton total reaction cross sections at 100 MeV *Can. J. Phys.* **44** 1847
- [78] Abegg R *et al* 1979 Measurement of the proton total reaction cross section for ^{159}Tb , ^{181}Ta and ^{197}Au between 20 and 48 MeV *Nucl. Phys. A* **324** 109
- [79] Auce A *et al* 2005 Reaction cross sections for protons on ^{12}C , ^{40}Ca , ^{90}Zr , and ^{208}Pb at energies between 80 and 180 MeV *Phys. Rev. C* **71** 064606
- [80] Smith A B 2005 Fast-neutrons incident on rotors: tantalum *Ann. Nucl. Energy* **32** 1926
- [81] Ferrer J C, Carlson J D and Rapaport J 1977 Neutron elastic scattering at 11 MeV and the isospin dependence of the neutron–nucleus optical potential *Nucl. Phys. A* **275** 325
- [82] Hansen L F, Dietrich F S, Pohl B A, Poppe C H and Wong C 1985 Test of microscopic optical model potentials for neutron elastic scattering at 14.6 MeV over a wide mass range *Phys. Rev. C* **31** 111
- [83] Greenlees G W, Poppe C H, Sievers J A and Watson D L 1971 Proton elastic scattering measurements at 9.8 MeV with optical-model analysis *Phys. Rev. C* **3** 1231
- [84] Gerstein G, Niederer J and Strauch K 1957 Elastic scattering of 96-MeV protons *Phys. Rev.* **108** 427
- [85] Steinberg D J, Palmieri J N and Cormack A M 1964 Small-angle scattering of 143 MeV polarized protons *Nucl. Phys.* **56** 46